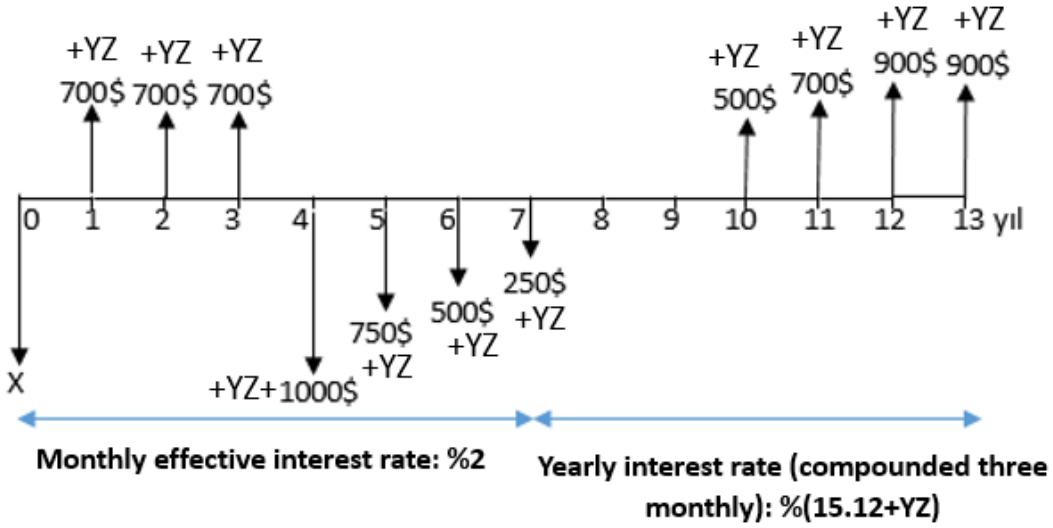
YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF CIVIL ENGINEERING
CONSTRUCTION MANAGEMENT
Final-Exam
14.06.2021
DURATION: 75'

NAME:								TOTAL POINTS	
SURNAME:									
STUDENT ID:				GROUP:					
Q1	Q2	Q3	-	-	-	-	-	-	-
			-	-	-	-	-	-	-

The term "XYZ" in the questions refers to the last three digits of your student ID.
Example: Student ID: 12396152 then the question will be solved by considering that "XYZ=152"

QUESTION 1. (30p) A firm should select one alternative among three alternatives to invest. The economic life of all alternatives is Y + 7 years. The initial cost of the first alternative is 150000 + XYZ, and the return in the first year is 50000, while this revenue from the second year will increase by 5000 + XYZ each year. The first cost of the second alternative is 180000. Its income in the first year is 70000. This income will decrease by 1500 + XYZ from the second year. The initial cost of the last alternative is 190000+ (10 * XYZ). Its income will remain constant at 60000 throughout its economic life. Within the framework of this principle, determine which of these alternatives to choose with the annual interest (Z + 8) % by using present value method. When Student No: 12396152, then XYZ is equal to "152" and the question should be solved in that way. **(Solve the question using continuously increasing or decreasing gradient series, solutions to be made with other methods will not be accepted.)**

QUESTION 2. (40p) For the cash flow shown, determine the present value.



QUESTION 3. (30p) For the cash flows shown, determine the future worth in year Z+8.

Year	Cash Flow $/Year	Estimated i Per year
0	5000+XYZ	(15+Z)%
1-(Z+4)	6000+100*(Y+Z)	(15+Z)%
Z+4- Z+8	9000	(18+Z)%

Single-Payment Compound-Amount Factor	**Single-Payment Present Worth Factor**	**Uniform Series Compound Amount**	**Sinking Fund Factor**
$F = P(1+i)^n = P(F/P,i,n)$	$P = F\left[\frac{1}{(1+i)^n}\right] = F(P/F,i,n)$	$F = A\left[\frac{(1+i)^n-1}{i}\right] = A(F/A,i,n)$	$A = F\left[\frac{i}{(1+i)^n-1}\right] = F(A/F,i,n)$
Capital Recovery Factor	**Uniform-Series Present Worth Factor**	**Uniform Gradient Series Factor**	**Uniform Gradient Series Factor**
$A = P\left[\frac{i(1+i)^n}{(1+i)^n-1}\right]$ $A = P(A/P,i,n)$	$P = A\left[\frac{(1+i)^n-1}{i(1+i)^n}\right]$ $P = A(P/A,i,n)$	$A = A_1 + G\left[\frac{1}{i} - \frac{n}{(1+i)^n-1}\right]$ $A = A_1 + G(A/G,i,n)$	$A = A_1 - G\left[\frac{1}{i} - \frac{n}{(1+i)^n-1}\right]$ $A = A_1 - G(A/G,i,n)$
Uniform Gradient Series for Feature Factor		$r = (1+\frac{j}{CK})^C - 1$	$r = (1+\frac{j}{m})^m - 1$
$F = Gx\frac{1}{i}\left[\frac{(1+i)^n-1}{i} - n\right] = G(F/G,i,n)$			